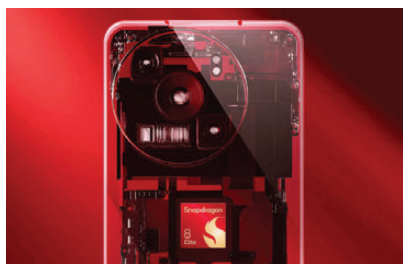


works in tandem with the Hexagon NPU to enhance every shot.

One of the standout features is the AI-powered Limitless Segmentation technology. This innovation can recognize and optimise over 250 layers within an image in real time. Whether it's adjusting the lighting on a subject's face, enhancing the colour of the sky, or sharpening details on distant objects, the technology ensures each element of the photo is captured in its best light. Computational photography enthusiasts are guaranteed to flip out!



For pet owners, the AI-based Pet Capture feature intelligently tracks and focuses on animals, ensuring that even the most energetic pets are captured with clarity. Additionally, the Video Object Eraser allows users to remove unwanted elements from video footage directly on the device — a tool that was once the purview of desktop editing software.

The camera system supports 4K video capture at 60 frames per second, even in low-light conditions, thanks to advanced noise reduction and image enhancement algorithms. With these capabilities, Qualcomm suggests users can produce professional-quality photos and videos without the need for external equipment.

Studio-grade audio with enhanced connectivity

Audio quality is often an overlooked aspect of the mobile experience, but the Snapdragon 8 Elite places it front and centre. With the Snapdragon Sound Technology Suite, Qualcomm claims users can enjoy lossless music streaming, ensuring every beat and note is reproduced true to source.

The inclusion of Qualcomm's Expanded Personal Area Network (XPAN) technology extends the range and bandwidth of wireless audio connections. This means you can move freely throughout a building without losing connection to your earbuds — a significant improvement over traditional Bluetooth limitations. For gamers and movie enthusiasts, lag-free spatial audio adds a new dimension to the experience, immersing the listener in a 3D soundscape that enhances engagement and enjoyment.

On the connectivity front, the Snapdragon 8 Elite is the first platform to integrate Wi-Fi, Bluetooth, and Ultra-Wideband (UWB) technologies. This integration enables hyper-accurate proximity sensing, allowing users to locate lost items, access secure buildings, or interact with smart home devices seamlessly.

The chipset features the Snapdragon X80 5G Modem-RF System combined with the Qualcomm 5G AI Suite Gen 3, which Qualcomm claims delivers superior 5G performance and multi-gigabit speeds. The Qualcomm FastConnect 7900 Mobile Connectivity System provides support for Wi-Fi 7, offering blazing-fast wireless connections and up to 40% power savings.

Snapdragon 8 Elite vs Competition

Early benchmarks and industry reports suggest that the Snapdragon 8 Elite doesn't just outperform its Android predecessors — it gives Apple's latest A18 Pro chip on the iPhone 16 Pro series a serious run for its money. This is particularly significant in an industry where Apple's silicon has long been considered the gold standard for mobile performance.

Manufactured on TSMC's advanced 3-nm process node, the Snapdragon 8 Elite benefits from increased transistor density, which translates to higher performance and improved energy efficiency. The use of Qualcomm's self-designed Oryon cores marks a strategic shift, allowing the company to

optimise the CPU architecture specifically for its target use cases.

The Adreno 830 GPU, with its tri-slice architecture running at 1.10 GHz, supports modern graphics APIs and advanced features like real-time ray tracing. Qualcomm claims a 35% improvement in ray tracing acceleration, enabling more realistic lighting and shadow effects in games and applications. Memory support has also been enhanced, with the chipset accommodating dual-channel LPDDR5X RAM running at 5.30 GHz. This ensures that the processor isn't bottlenecked by slower memory speeds, maintaining smooth performance even under heavy multitasking or gaming loads.

The Snapdragon 8 Elite is set to power a wave of flagship Android smartphones from leading manufacturers like Samsung, Xiaomi, and OnePlus. Devices featuring this chipset are expected to hit the market between October and December 2024, with the Samsung Galaxy S25 series among the anticipated launches.

For consumers, this means a new generation of smartphones that not only push the envelope in terms of raw performance but also offer enhanced AI capabilities, superior gaming experiences, and advanced camera functionalities. Qualcomm's emphasis on AI integration across all aspects of the chipset — from the CPU and GPU to the ISP and connectivity modules — signals a shift toward more intelligent, context-aware devices. As AI models become more sophisticated, on-device processing will be crucial for privacy, efficiency, and responsiveness.

As we move into 2025, the impact of the Snapdragon 8 Elite will be felt across the Android ecosystem. It raises the bar for competitors and promises consumers devices that are not just smarter and faster but also more attuned to their needs and lifestyles. For those contemplating their next smartphone upgrade, the arrival of Snapdragon 8 Elite-powered devices is an exciting development worth keeping an eye on. **d**